IV Tutorial

Rolling CAPM, Okun's law and structural breaks

Exercise 1. Rolling CAPM and parameter stability

Notebook: Rolling_CAPM.ipynb

Analyse IBM's monthly excess returns and the broad US market excess return for 2000–2025 (312 months). Compare full-sample CAPM estimates with 60-month rolling regressions. Interpret alpha and beta, their uncertainty and changes across windows.

Relevant concepts

- Simple and excess returns; alpha; market beta; coefficient units.
- Standard errors; zero-coefficient tests; p-values; confidence intervals; R^2.
- Full-sample versus rolling estimates; window length; precision and recency.
- Parameter stability; pointwise confidence intervals; overlapping windows; sampling variation versus structural change.

To do. Calculate the fitted IBM excess return when the market excess return is 1%, using first the full-sample coefficients and then those for the window ending in December 2025. Interpret the difference and comment on the rolling beta plot. Explain why beta equal to 1 means market-like sensitivity and why a changing estimate alone does not establish a structural break.

Exercise 2. Asymptotic bias: ignoring structural breaks

Let r_t be serially uncorrelated returns, with $\mathrm{Var}(r_t) = 1$. The expected return changes once, after a fraction $\tau \in (0, 1)$ of the sample:

$$E[r_t] = a + b\,\mathbf{1}\{t > k_T\}, \qquad k_T = \lfloor \tau T \rfloor, \qquad t = 1, \ldots, T.$$

Here k_T is the number of pre-break observations, a is the pre-break mean and $a + b$ is the post-break mean. Ignore the break and estimate by OLS a model with only a constant:

$$r_t = \beta_0 + u_t.$$

To do.

(a) Derive $\widehat{\beta}_0$ and its probability limit as $T \to \infty$. Show how the pre-break and post-break sample means enter the estimator.

(b) Interpret the limit and its weights. Does using more observations recover either regime's mean when $b \neq 0$?

(c) Suppose the post-break regime continues after T. Find the limiting bias of the constant forecast relative to the future expected return, and explain the implication for forecasting.

Exercise 3. Okun's law: growth and unemployment

Notebook: Rolling_Regression_Parameter_Stability.ipynb

Analyse quarterly US real GDP growth and changes in unemployment for 1980Q1–2025Q3 (183 quarters). Compare a full-sample linear OLS regression with 40-quarter rolling estimates. Interpret the coefficients and their uncertainty, considering parameter stability and possible endogeneity.

Relevant concepts

- Growth rates versus levels; percentage changes versus percentage points.
- Linear OLS; intercept and slope; fitted unemployment changes; coefficient units.
- HAC standard errors; confidence intervals; influential observations.
- Exogeneity; simultaneity; omitted variables; limits of HAC inference.
- Full-sample versus rolling estimates; window length; sampling variation versus parameter instability.

Exercise 4. Case study: a bank during financial stress

An analyst has monthly returns for a listed bank, the broad stock market and a matching risk-free investment, from January 2005 to December 2015. All returns are measured in percent.

The analyst wants to examine whether the bank's average return relative to the risk-free investment was different during September 2008–June 2009, after accounting for market movements. For this case, those months define the period of financial stress. Assume the bank's sensitivity to the market is the same throughout the sample.

To do. Propose an OLS model for this question. Write the equation and clearly define its dependent variable and each regressor. No estimation or numerical calculations are required.

Solutions

Exercise 2. Asymptotic bias: ignoring structural breaks

(a) OLS estimator and probability limit

OLS minimises $\sum_{t=1}^{T}(r_t - \beta_0)^2$, giving

$$\widehat{\beta}_0 = \frac{1}{T}\sum_{t=1}^{T} r_t = \bar{r}.$$

Define the two regime-specific sample means:

$$\bar{r}_{\text{pre}} = \frac{1}{k_T}\sum_{t=1}^{k_T} r_t, \qquad \bar{r}_{\text{post}} = \frac{1}{T-k_T}\sum_{t=k_T+1}^{T} r_t.$$

Splitting the sum and inserting each regime's sample size,

$$\begin{aligned}\widehat{\beta}_0 &= \frac{1}{T}\sum_{t=1}^{k_T} r_t + \frac{1}{T}\sum_{t=k_T+1}^{T} r_t \\ &= \frac{k_T}{T}\bar{r}_{\text{pre}} + \left(1 - \frac{k_T}{T}\right)\bar{r}_{\text{post}}.\end{aligned}$$

The pre-break mean has expectation a and variance $1/k_T$; the post-break mean has expectation $a+b$ and variance $1/(T-k_T)$. Serial uncorrelatedness is sufficient for these variance expressions. Both regime sample sizes grow, so their variances vanish. Chebyshev's inequality gives $\bar{r}_{\text{pre}} \xrightarrow{p} a$ and $\bar{r}_{\text{post}} \xrightarrow{p} a+b$. Also $k_T/T \to \tau$. Therefore,

$$\boxed{\operatorname{plim}\widehat{\beta}_0 = \tau a + (1-\tau)(a+b) = a + (1-\tau)b.}$$

(b) Interpretation

The limit is a **weighted average of the two regime means**: τ is the pre-break share and $1-\tau$ is the post-break share. For $b \neq 0$, it lies strictly between a and $a+b$. More observations reduce sampling uncertainty ($\operatorname{Var}(\widehat{\beta}_0) = 1/T$), but do not recover either regime's mean. This limit keeps τ fixed, so both regimes retain a positive sample share.

(c) Forecasting after the break

If the post-break regime persists, the future expected return is $a+b$. The limiting forecast bias, defined as forecast minus true expected return, is

$$\boxed{\operatorname{plim}\widehat{\beta}_0 - (a+b) = a + (1-\tau)b - (a+b) = -\tau b.}$$

An upward shift ($b > 0$) leads to underprediction; a downward shift ($b < 0$) leads to overprediction. More data make the estimate more precise without removing this bias relative to the current regime.

Exercise 4. Case study: a bank during financial stress

One suitable model is

$$R_{B,t} - R_{f,t} = \beta_0 + \beta_1(R_{m,t} - R_{f,t}) + \beta_2 C_t + u_t, \qquad C_t = \begin{cases} 1, & \text{September 2008–June 2009,} \\ 0, & \text{otherwise.} \end{cases}$$

The dependent variable is the bank's excess return, $R_{B,t} - R_{f,t}$. The regressors are a constant, the market excess return $R_{m,t} - R_{f,t}$, and a **dummy variable** C_t identifying the stress period.

At a given market excess return, the fitted bank return during stress differs by β_2 percentage points. The market slope β_1 is common to both periods, as stipulated. The dummy allows the intercept to change; its coefficient describes a conditional association, rather than an identified causal effect of the crisis.